UTS Praktek Pemrograman Klien Server



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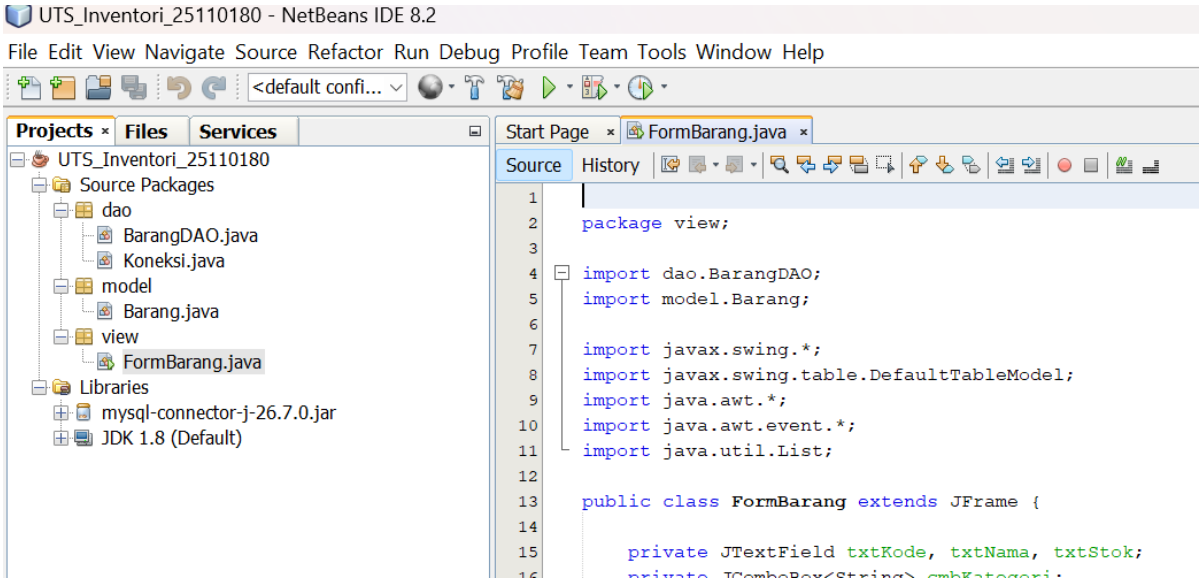
Nim : 25110180

Kelas : TK – c

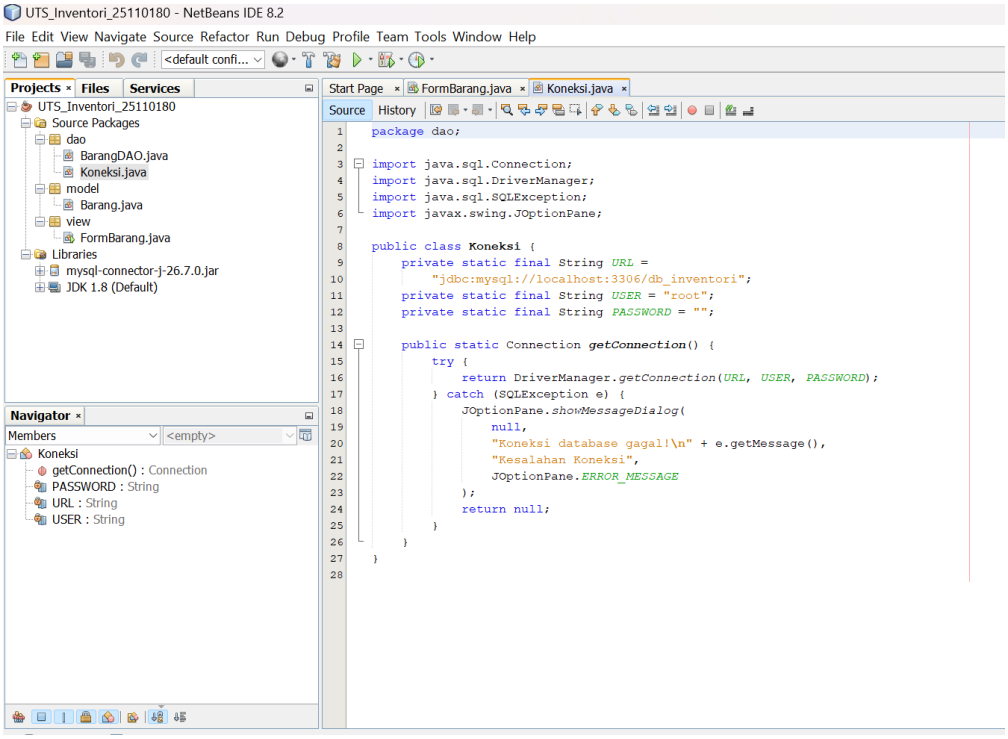
POLITEKNIK BISNIS INDONESIA

A. Tugas

1. Proyek Netbeans



2. Class Koneksi



3. Class model Barang

```java

package model;

public class Barang {
    private String kodeBarang;
    private String namaBarang;
    private String kategori;
    private int stok;

    public Barang() {
    }

    public Barang(String kodeBarang, String namaBarang,
                  String kategori, int stok) {
        this.kodeBarang = kodeBarang;
        this.namaBarang = namaBarang;
        this.kategori = kategori;
        this.stok = stok;
    }

    public String getKodeBarang() {
        return kodeBarang;
    }

    public void setKodeBarang(String kodeBarang) {
        this.kodeBarang = kodeBarang;
    }

    public String getNamaBarang() {
        return namaBarang;
    }

    public void setNamaBarang(String namaBarang) {
        this.namaBarang = namaBarang;
    }

    public String getKategori() {
        return kategori;
    }

    public void setKategori(String kategori) {
        this.kategori = kategori;
    }

    public int getStok() {
        return stok;
    }

    public void setStok(int stok) {
        this.stok = stok;
    }
}

```

4. FormBarang
   Coding :

```java

package view;

import dao.BarangDAO;
import model.Barang;

import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.*;
import java.util.List;

public class FormBarang extends JFrame {

    private JTextField txtKode, txtNama, txtStok;
    private JComboBox<String> cmbKategori;
    private JTable tabel;
    private DefaultTableModel model;

    private final BarangDAO dao = new BarangDAO();

    public FormBarang() {
        setTitle("Sistem Inventori Barang Laboratorium");
        setSize(800, 550);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        buatForm();
        tampilkanData();
    }

```

```java
private void buatForm() {
    JPanel utama = new JPanel(new BorderLayout(10, 10));
    utama.setBorder(BorderFactory.createEmptyBorder(15, 15, 15, 15));
    utama.setBackground(new Color(245, 248, 252));

    JLabel judul = new JLabel("DATA BARANG LABORATORIUM");
    judul.setFont(new Font("Arial", Font.BOLD, 22));
    judul.setForeground(new Color(30, 70, 120));

    JPanel panelAtas = new JPanel(new BorderLayout());
    panelAtas.setOpaque(false);
    panelAtas.add(judul, BorderLayout.CENTER);

    JPanel panelInput = new JPanel(new GridLayout(4, 2, 10, 10));
    panelInput.setBorder(BorderFactory.createTitledBorder("Form Barang"));
    panelInput.setBackground(Color.WHITE);

    txtKode = new JTextField();
    txtNama = new JTextField();
    txtStok = new JTextField();

    cmbKategori = new JComboBox<>(new String[]{
        "Komputer", "Jaringan", "Alat Ukur"
    });

    panelInput.add(new JLabel("Kode Barang"));
    panelInput.add(txtKode);
    panelInput.add(new JLabel("Nama Barang"));
    panelInput.add(txtNama);
    panelInput.add(new JLabel("Kategori"));
    panelInput.add(cmbKategori);
    panelInput.add(new JLabel("Stok"));
    panelInput.add(txtStok);
```

```java
    JButton btnSimpan = new JButton("Simpan");
    JButton btnUbah = new JButton("Ubah");
    JButton btnHapus = new JButton("Hapus");
    JButton btnBersihkan = new JButton("Bersihkan");

    JPanel panelTombol = new JPanel(new FlowLayout(FlowLayout.LEFT));
    panelTombol.setOpaque(false);
    panelTombol.add(btnSimpan);
    panelTombol.add(btnUbah);
    panelTombol.add(btnHapus);
    panelTombol.add(btnBersihkan);

    JPanel panelForm = new JPanel(new BorderLayout(5, 5));
    panelForm.setOpaque(false);
    panelForm.add(panelInput, BorderLayout.CENTER);
    panelForm.add(panelTombol, BorderLayout.SOUTH);

    model = new DefaultTableModel(
        new String[]{"Kode", "Nama Barang", "Kategori", "Stok"}, 0
    ) {
        @Override
        public boolean isCellEditable(int row, int column) {
            return false;
        }
    };

    tabel = new JTable(model);
    tabel.setSelectionMode(ListSelectionModel.SINGLE_SELECTION);
    tabel.setRowHeight(24);

    JScrollPane scroll = new JScrollPane(tabel);
    scroll.setBorder(BorderFactory.createTitledBorder("Daftar Barang"));
```

```java
    utama.add(panelAtas, BorderLayout.NORTH);
    utama.add(panelForm, BorderLayout.CENTER);
    utama.add(scroll, BorderLayout.SOUTH);

    // Mengatur tinggi area tabel
    scroll.setPreferredSize(new Dimension(750, 220));

    add(utama);

    btnSimpan.addActionListener(e -> simpanData());
    btnUbah.addActionListener(e -> ubahData());
    btnHapus.addActionListener(e -> hapusData());
    btnBersihkan.addActionListener(e -> bersihkanForm());

    tabel.addMouseListener(new MouseAdapter() {
        @Override
        public void mouseClicked(MouseEvent e) {
            int baris = tabel.getSelectedRow();

            if (baris >= 0) {
                txtKode.setText(model.getValueAt(baris, 0).toString());
                txtNama.setText(model.getValueAt(baris, 1).toString());
                cmbKategori.setSelectedItem(
                    model.getValueAt(baris, 2).toString()
                );
                txtStok.setText(model.getValueAt(baris, 3).toString());

                txtKode.setEditable(false);
            }
        }
    });
}
```

```java
    private Barang ambilDataForm() {
        String kode = txtKode.getText().trim();
        String nama = txtNama.getText().trim();
        String stokText = txtStok.getText().trim();
        String kategori = cmbKategori.getSelectedItem().toString();

        if (kode.isEmpty() || nama.isEmpty()) {
            JOptionPane.showMessageDialog(this,
                "Kode dan nama barang tidak boleh kosong!");
            return null;
        }

        if (kode.length() > 10) {
            JOptionPane.showMessageDialog(this,
                "Kode barang maksimal 10 karakter!");
            return null;
        }

        int stok;
        try {
            stok = Integer.parseInt(stokText);
            if (stok < 0) {
                JOptionPane.showMessageDialog(this,
                    "Stok tidak boleh kurang dari 0!");
                return null;
            }
        } catch (NumberFormatException e) {
            JOptionPane.showMessageDialog(this,
                "Stok harus berupa angka bulat >= 0!");
            return null;
        }

        return new Barang(kode, nama, kategori, stok);
    }

    private void simpanData() {
        Barang b = ambilDataForm();
        if (b == null) return;

        try {
            if (dao.simpan(b)) {
                JOptionPane.showMessageDialog(this,
                    "Data barang berhasil disimpan!");
                tampilkanData();
                bersihkanForm();
            } else {
                JOptionPane.showMessageDialog(this,
                    "Data gagal disimpan.");
            }
        } catch (RuntimeException e) {
            JOptionPane.showMessageDialog(this,
                "Gagal menyimpan data: " + e.getMessage());
        }
    }

    private void ubahData() {
        Barang b = ambilDataForm();
        if (b == null) return;

        try {
            if (dao.ubah(b)) {
                JOptionPane.showMessageDialog(this,
                    "Data barang berhasil diubah!");
                tampilkanData();
                bersihkanForm();
            } else {
                JOptionPane.showMessageDialog(this,
                    "Data tidak ditemukan atau tidak berubah.");
            }
        } catch (RuntimeException e) {
            JOptionPane.showMessageDialog(this,
                "Gagal mengubah data: " + e.getMessage());
        }
    }

    private void hapusData() {
        String kode = txtKode.getText().trim();

        if (kode.isEmpty()) {
            JOptionPane.showMessageDialog(this,
                "Pilih barang yang akan dihapus!");
            return;
        }

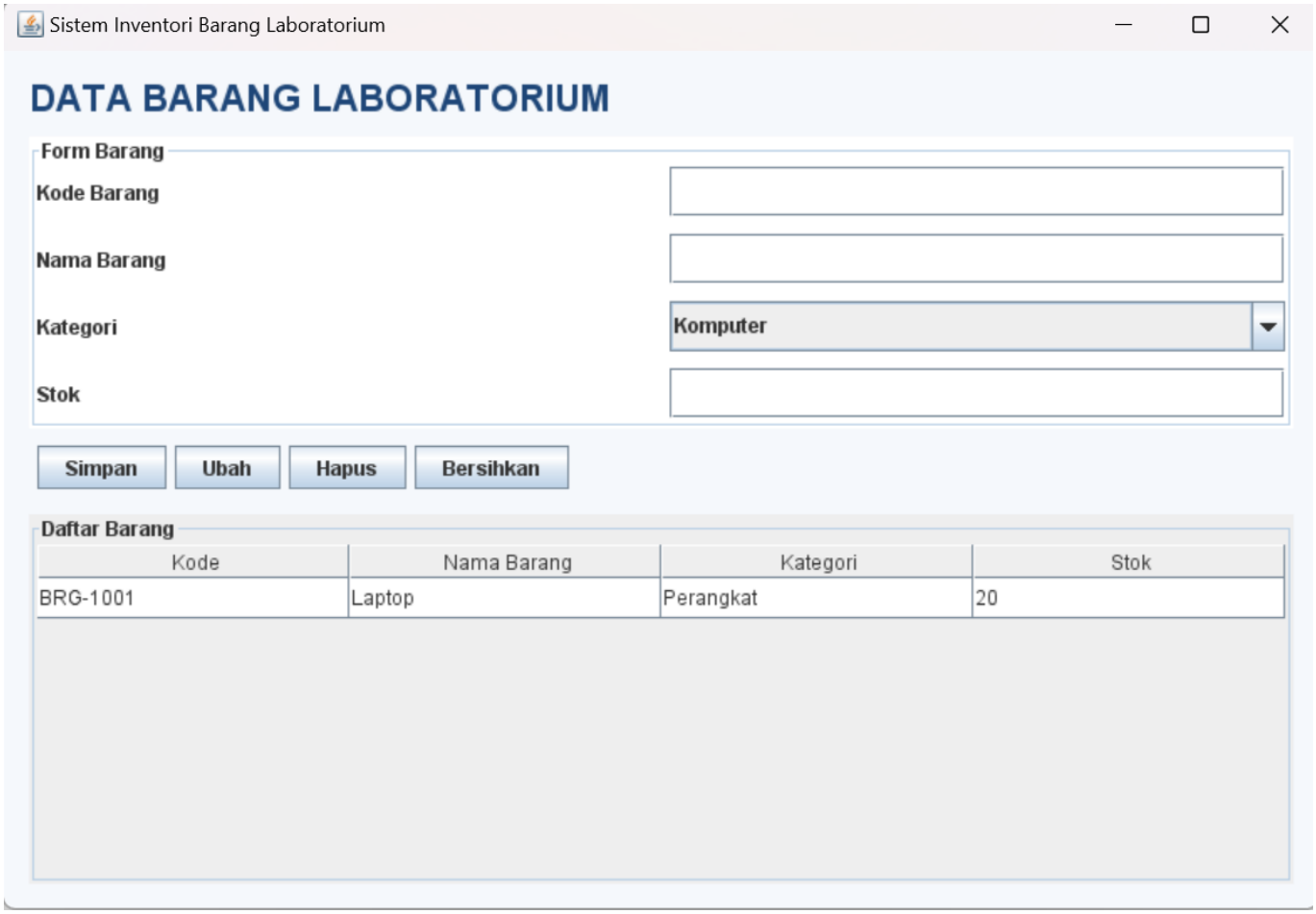
        int konfirmasi = JOptionPane.showConfirmDialog(
            this,
            "Hapus barang dengan kode " + kode + "?",
            "Konfirmasi Hapus",
            JOptionPane.YES_NO_OPTION
        );

        if (konfirmasi != JOptionPane.YES_OPTION) return;

        try {
            if (dao.hapus(kode)) {
                JOptionPane.showMessageDialog(this,
                    "Data barang berhasil dihapus!");
                tampilkanData();
                bersihkanForm();
            } else {
                JOptionPane.showMessageDialog(this,
                    "Data tidak ditemukan.");
            }
        } catch (RuntimeException e) {
            JOptionPane.showMessageDialog(this,
                "Gagal menghapus data: " + e.getMessage());
        }
    }
```

```
241     private void tampilkanData() {
242         model.setRowCount(0);
243 
244         try {
245             List<Barang> daftar = dao.getAll();
246 
247             for (Barang b : daftar) {
248                 model.addRow(new Object[]{
249                     b.getKodeBarang(),
250                     b.getNamaBarang(),
251                     b.getKategori(),
252                     b.getStok()
253                 });
254             }
255         } catch (RuntimeException e) {
256             JOptionPane.showMessageDialog(this,
257                 "Gagal menampilkan data: " + e.getMessage());
258         }
259     }
260 
261     private void bersihkanForm() {
262         txtKode.setText("");
263         txtNama.setText("");
264         txtStok.setText("");
265         cmbKategori.setSelectedIndex(0);
266         txtKode.setEditable(true);
267         tabel.clearSelection();
268         txtKode.requestFocus();
269     }
270 
271     public static void main(String[] args) {
272         SwingUtilities.invokeLater(() -> {
273             new FormBarang().setVisible(true);
274         });
275     }
276 }
277 
```

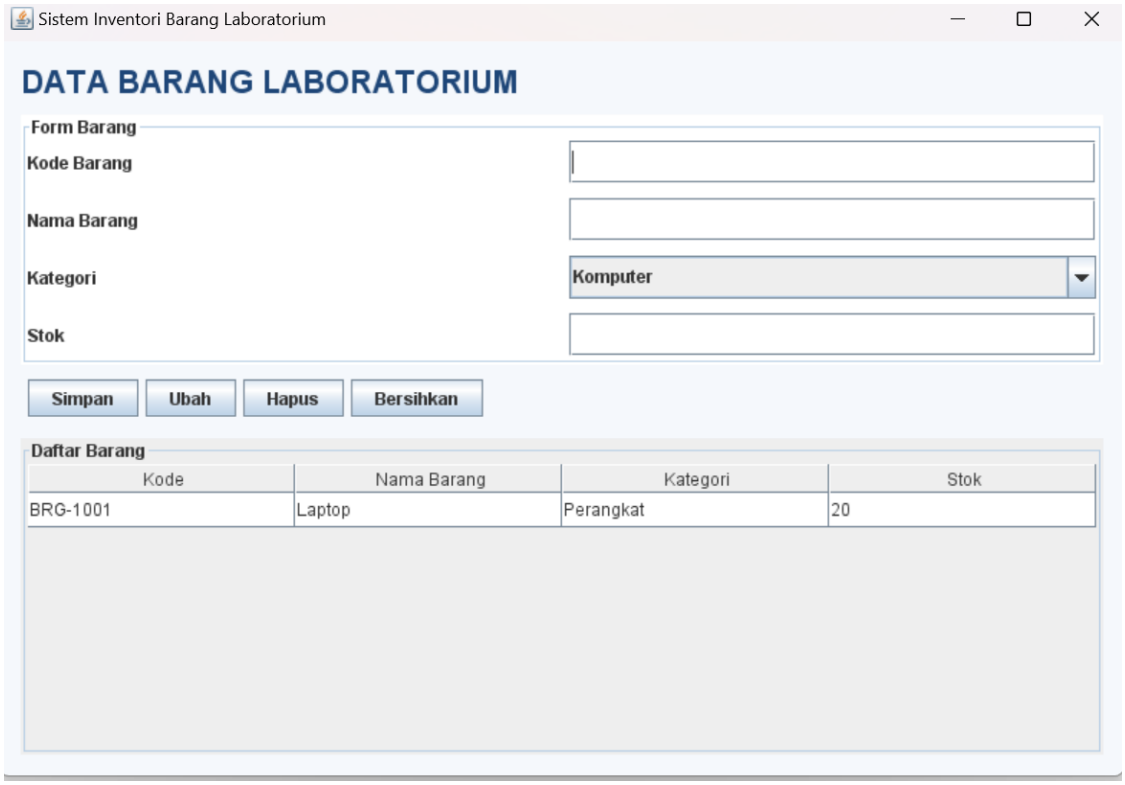
Tampilan :



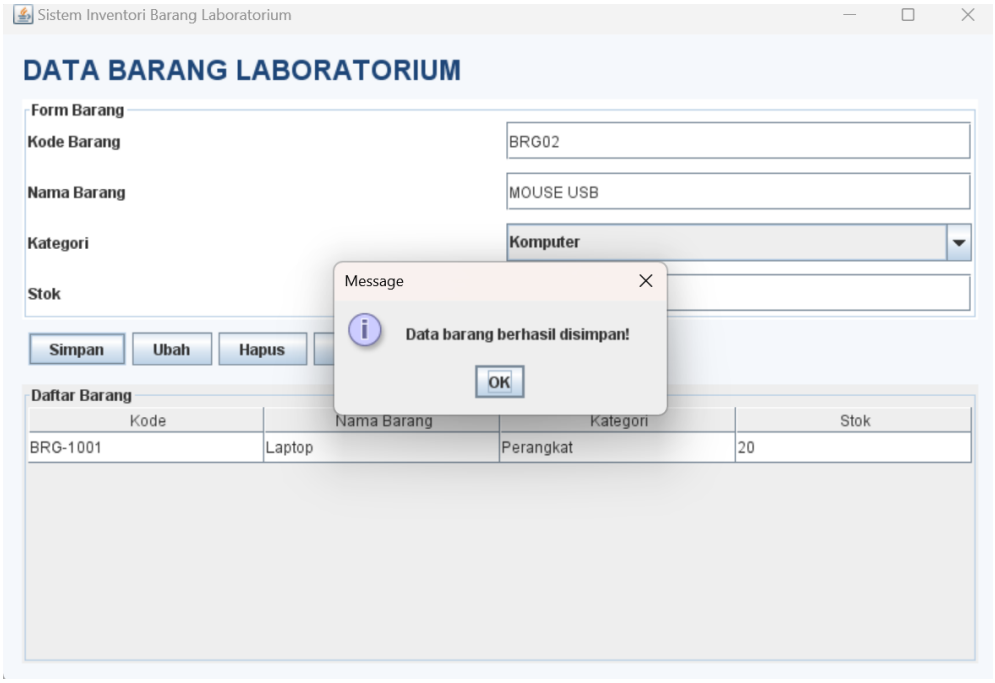
5. Tombol Simpan

```
167     private void simpanData() {
168         Barang b = ambilDataForm();
169         if (b == null) return;
170 
171         try {
172             if (dao.simpan(b)) {
173                 JOptionPane.showMessageDialog(this,
174                     "Data barang berhasil disimpan!");
175                 tampilkanData();
176                 bersihkanForm();
177             } else {
178                 JOptionPane.showMessageDialog(this,
179                     "Data gagal disimpan.");
180             }
181         } catch (RuntimeException e) {
182             JOptionPane.showMessageDialog(this,
183                 "Gagal menyimpan data: " + e.getMessage());
184         }
185     }
186 
```

6. Data baru dibuka



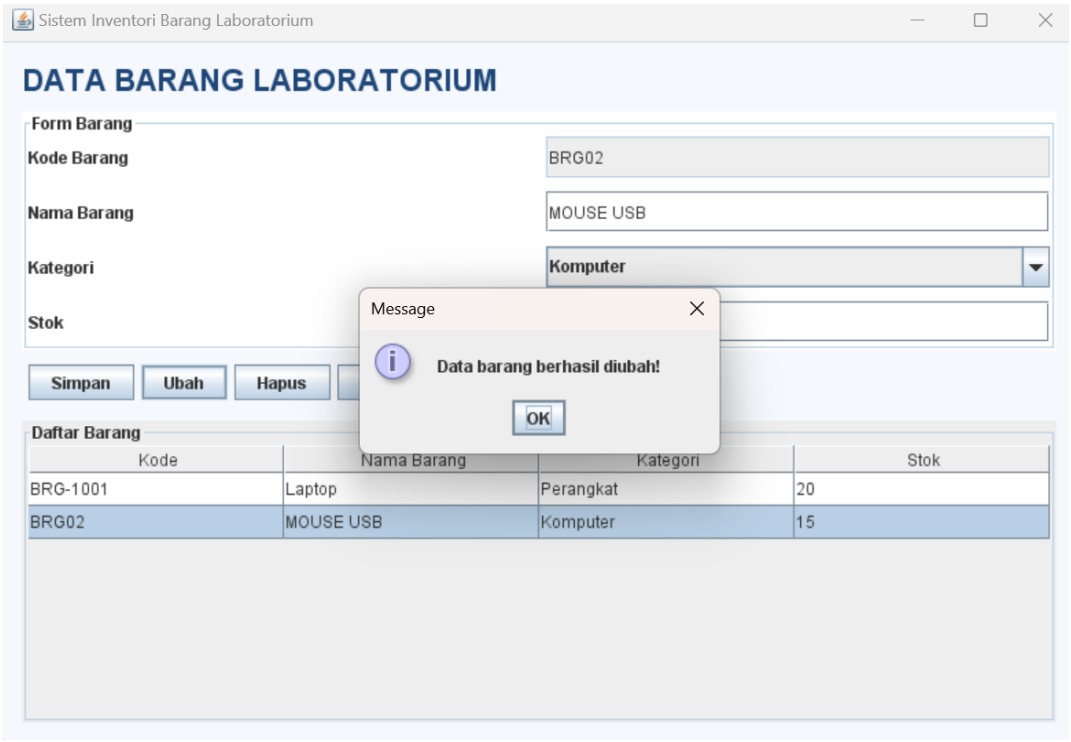
Data disimpan :



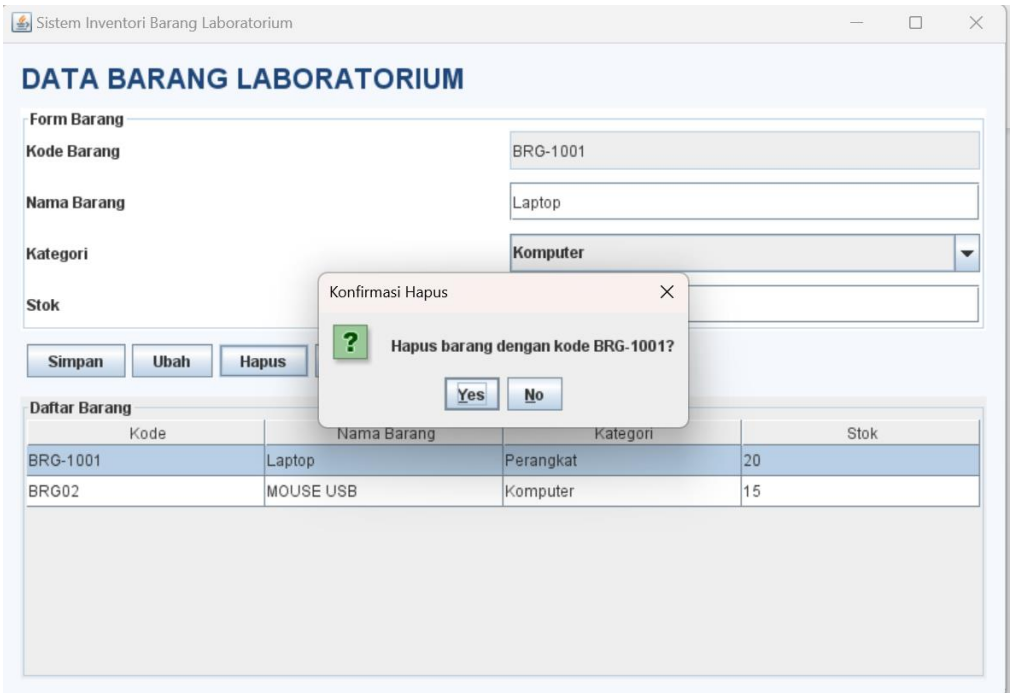
Data barang saat di klik :

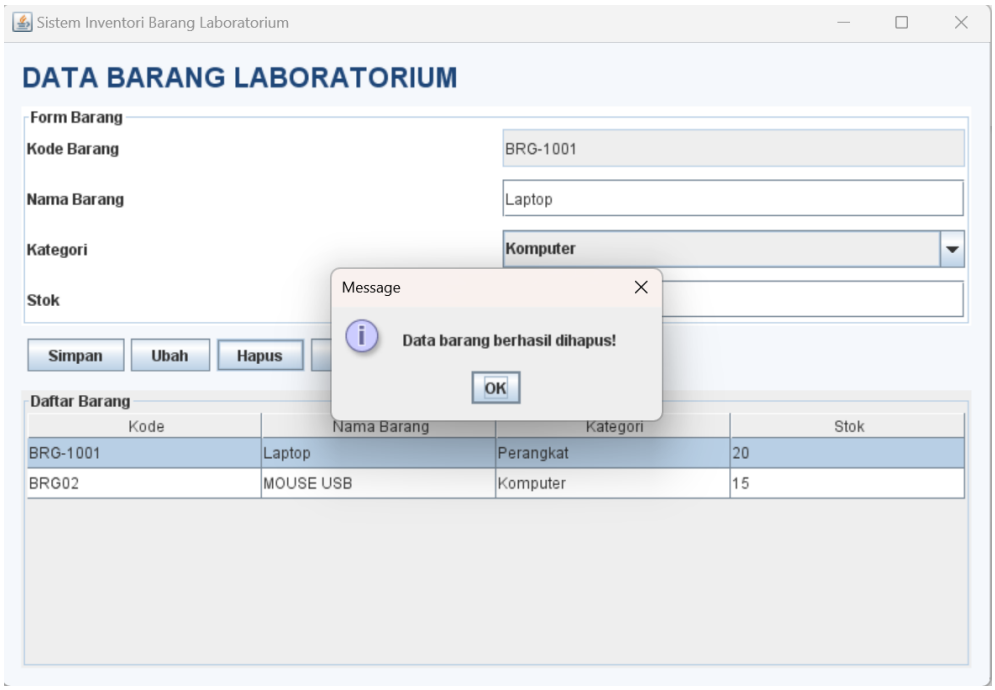


7. Tombol ubah dan hapus
   Ubah :

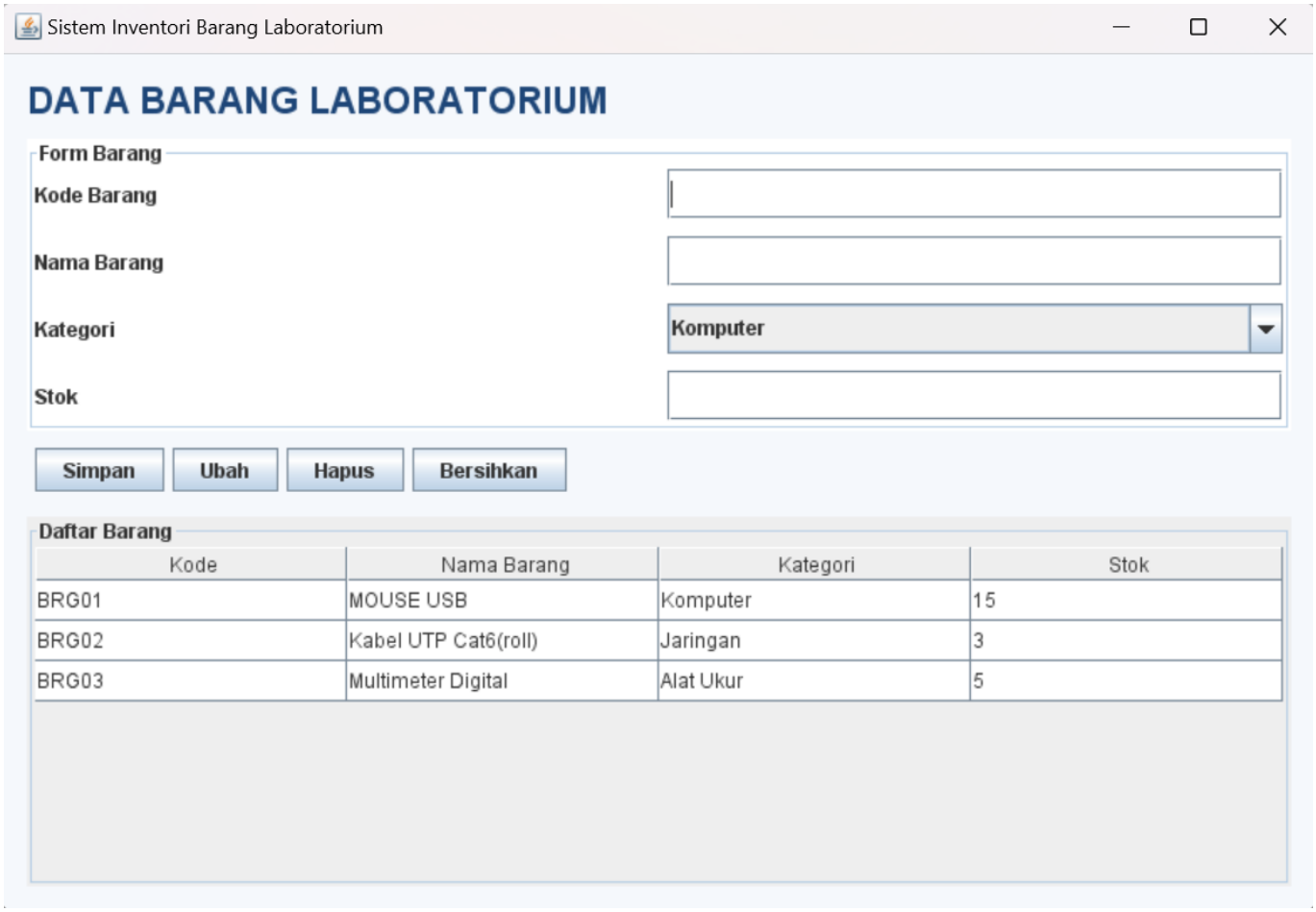


   Hapus :

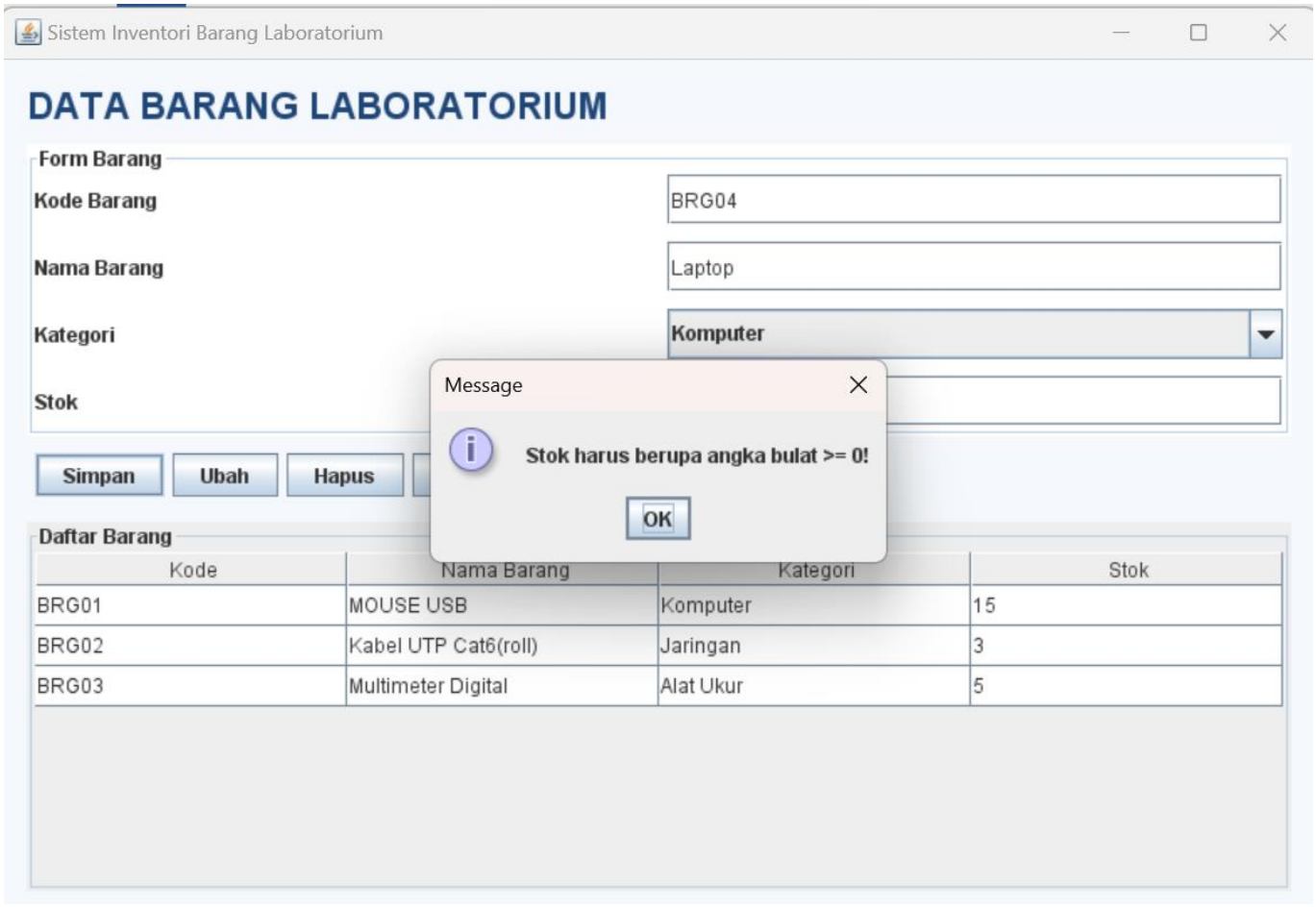



Sistem Inventori Barang Laboratorium
DATA BARANG LABORATORIUM
Form Barang
Kode Barang
BRG-1001
Nama Barang
Laptop
Kategori
Komputer
Stok
Message
Data barang berhasil dihapus!
OK
Simpan
Ubah
Hapus
Daftar Barang
Kode
Nama Barang
Kategori
Stok
BRG-1001
Laptop
Perangkat
20
BRG02
MOUSE USB
Komputer
15

Data Uji :


Sistem Inventori Barang Laboratorium
DATA BARANG LABORATORIUM
Form Barang
Kode Barang
Nama Barang
Kategori
Komputer
Stok
Simpan
Ubah
Hapus
Bersihkan
Daftar Barang
Kode
Nama Barang
Kategori
Stok
BRG01
MOUSE USB
Komputer
15
BRG02
Kabel UTP Cat6(roll)
Jaringan
3
BRG03
Multimeter Digital
Alat Ukur
5

Input Salah :



Database

